1 Visual region extraction

Patch–box intersection-area weights
224 × 224 · no CLS token



Input image I

CLIP ViT-B/16 ❄
visual encoder

Patch features
14 × 14 × 768

YOLO detector ❄

Boxes B
(x_1, y_1, x_2, y_2)

Box-weighted
ROI pooling

Region descriptors
N × 768

Linear projection 🔥
768 → 512

V

2 Semantic bank–region alignment

P_n · 150 × 512
A

V_1, T_1

Object fusion 1 🔥
B

Object fusion 2 🔥
B

V_o

V : N × 512

P_v · 150 × 512
A

Relation fusion 🔥
B

V_r

A TRAIN-only bank preparation & aggregation

B One shared fusion detail; separate parameters

3 Image-specific prompt–region alignment

UPT boxes/classes + YOLO boxes/classes

Class-consistent matching · IoU ≥ 0.5
ROI index i / ordered pair (s, o)

Unmatched slots are omitted.

t_o / t_r + ROI links feed the corresponding Slot MLP.

Object adapter · C 🔥
t_o + linked ROI metadata

Relation adapter · C 🔥
t_r + linked endpoint metadata

I → UPT detector ❄
same input image

Object / relation
phrases

CLIP text encoder ❄
t_o / t_r

Named ports → C

V_o'

V_r'

4 Caption generation

Parallel decoder · D 🔥
V_o' and V_r' remain separate
3 layers · dimension 512 · heads 8
Prefix → next-token distribution

Word embedding + fixed PE

5 Caption reranking

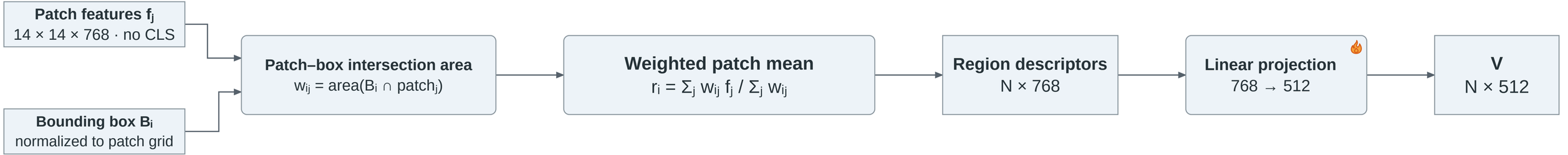
Beam search · width 5
max 20 new tokens

Five candidate captions

Three-score reranking · E
S = 0.3D + 0.4C + 0.3O

Final caption

Visual detail Box-weighted ROI pooling



One weighted descriptor per box; partial patch overlaps contribute proportionally. No crop image encoder.

A TRAIN banks & semantic aggregation

TRAIN captions
train + restval

spaCy POS
+ lemmatization

Noun / verb
lemmas

CLIP text
encoder

OFFLINE · BUILT ONCE

TRAIN references only

No VALID / TEST references

Shared banks ≠ image-specific prompts

150 soft prototypes per bank;
not 150 selected words.

Fixed noun bank E_n
512-D embeddings

Fixed verb bank E_v
512-D embeddings

Learned semantic
aggregation

Learned semantic
aggregation

P_n
150 × 512

P_v
150 × 512

Learned aggregation · separate noun / verb parameters

E

Linear W_a

MLP

Softmax
over bank entries

Weights A

Linear W_b

Transformed embeddings
E W_b

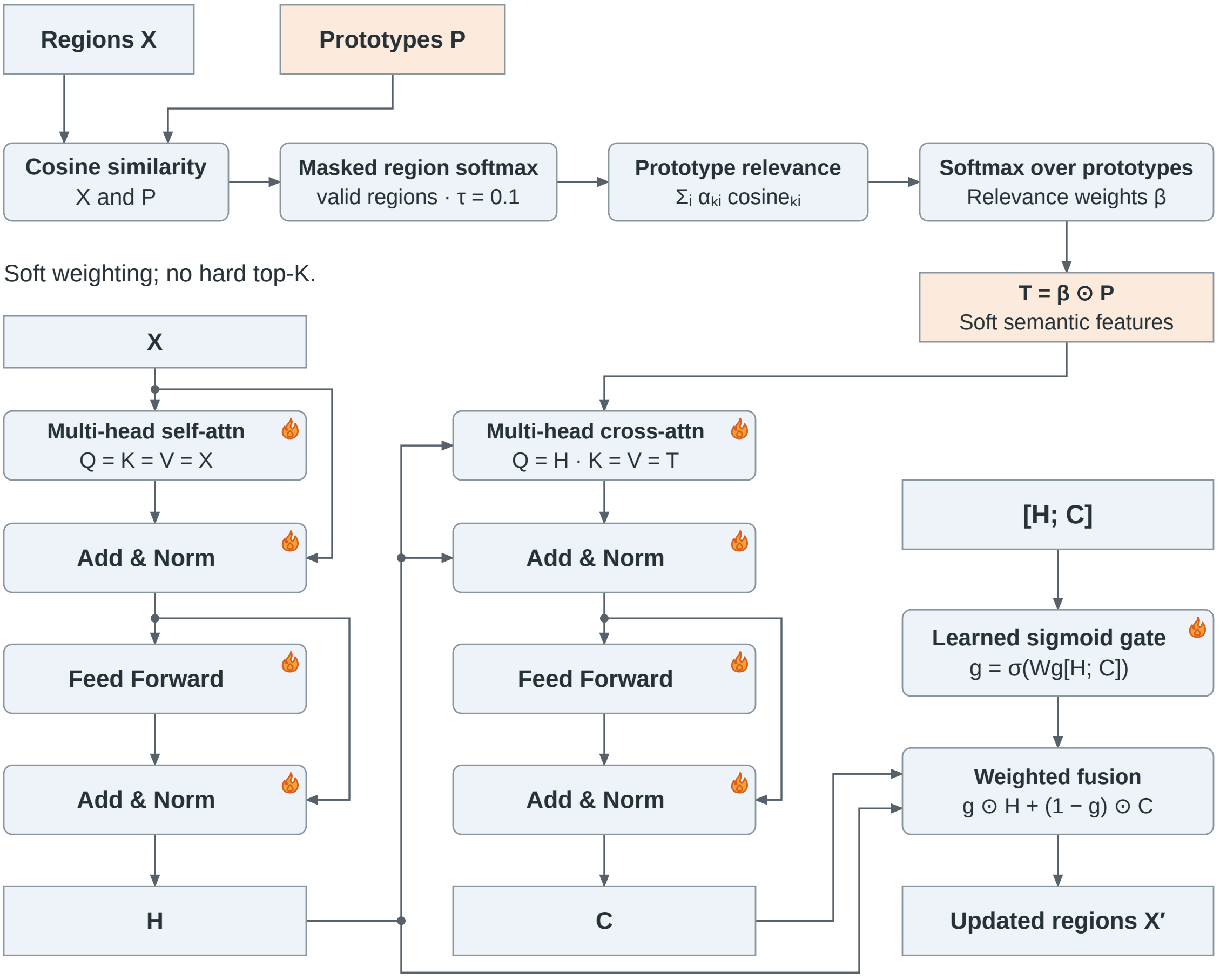
A^T × E W_b

P : 150 × 512

$$A = \text{softmax_bank}(\text{MLP}(EW_a))\ P = A^T(EW_b)$$

B Semantic fusion block

Same structure, separate parameters · two outputs: X′ and T

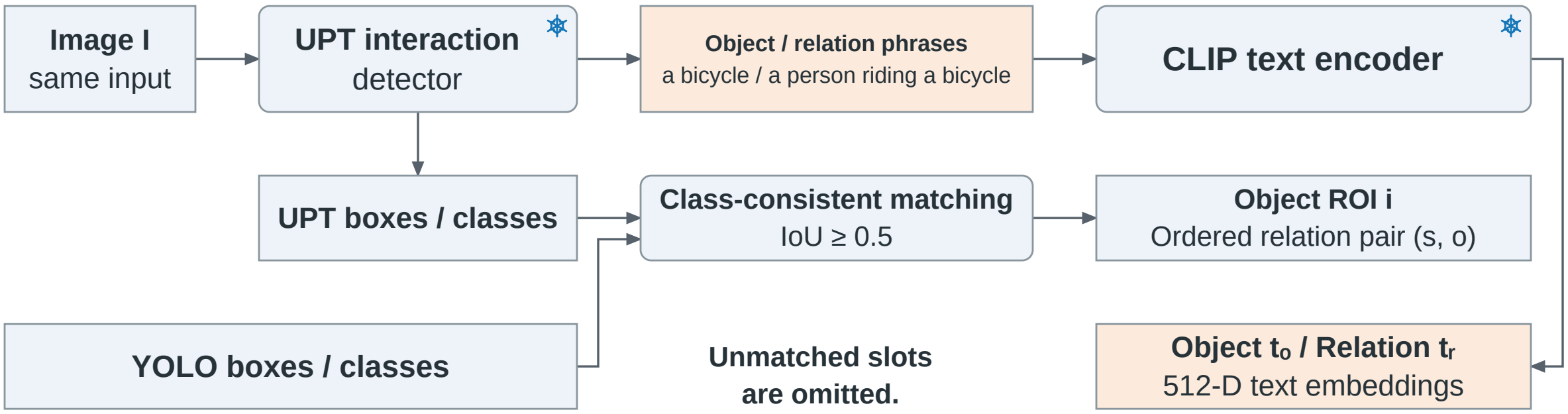


Soft weighting; no hard top-K.

Object fusion 1: $(X, P) = (V, P_n) \rightarrow (V_1, T_1)$

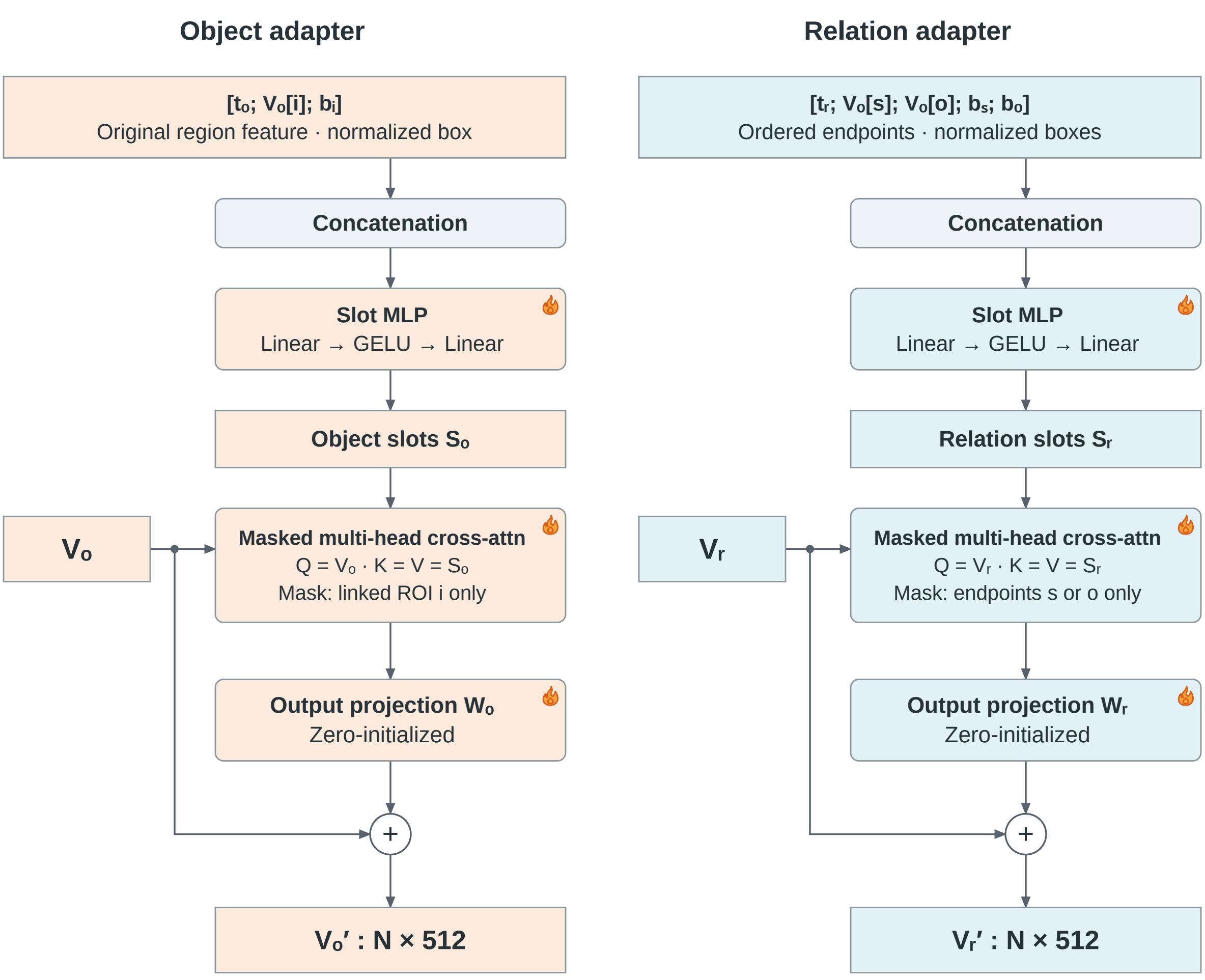
Object fusion 2: $(V_1, T_1) \rightarrow V_o$ Relation fusion: $(V_o, P_v) \rightarrow V_r$

C Image-specific prompt extraction & ROI matching



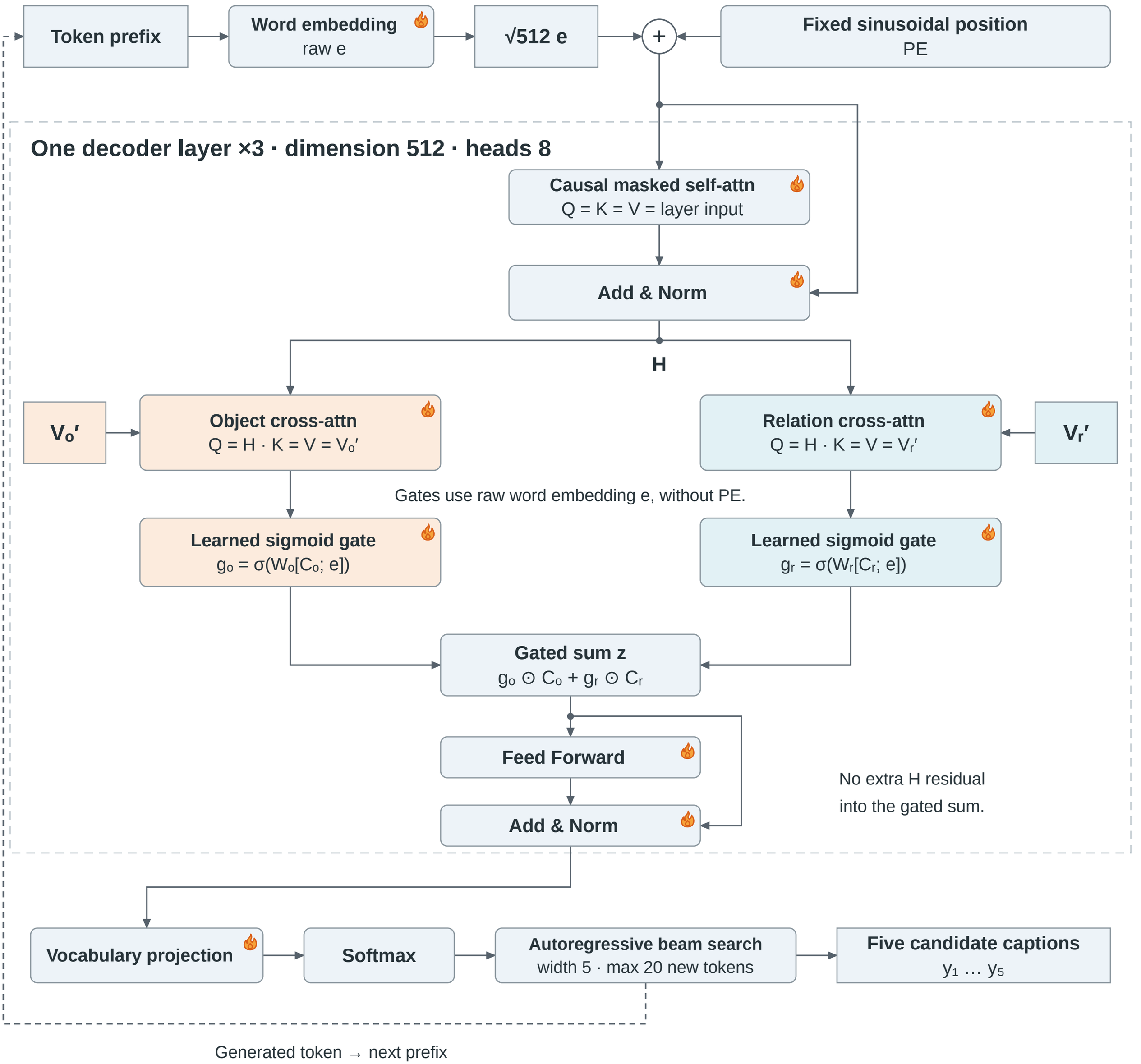
C Two image-specific prompt adapters

Both slot builders use original V_o. [;] denotes concatenation.



No valid linked slot → zero residual. No adapter gate.

D Caption generation



E Caption reranking

Same five candidates $y_1 \dots y_5$

D · Decoder likelihood

Sum token log-probabilities
Raw sequence log-probability

Min–max normalization
over the same 5 candidates

No length normalization.

D

C · Image–text alignment

CLIP image / text encoders
Image I + candidate caption

Cosine-based CLIPScore
2.5 max(cosine, 0)

Min–max normalization
over the same 5 candidates

C

O · Object consistency

Mentioned object extraction
from candidate caption

Supported object labels
from YOLO detections

Detector-based
object consistency

Normalized object score
(consistency + 1) / 2

O

$$S(y) = 0.3\,D(y) + 0.4\,C(y) + 0.3\,O(y)$$

Select highest-scoring candidate

Final caption

Equal scores: earlier beam wins.

References and CHAIR are not used for selection.

F Training only

Stage 1 · Cross-entropy training

TRAIN reference caption

Shift right

Current decoder

Predicted token distribution

Target tokens

LCE

Stage 2 · SCST + cross-entropy

Current model · same image

Sampled caption y_s

Greedy caption yg

TRAIN references

CIDEr reward r(·)

Current-batch LCE
TRAIN teacher forcing

Advantage: $r(y_s) - r(yg)$

Rollout log $p(y_s)$

LSCST · detached advantage

Total loss = LCE + 0.05 LSCST

References: training and evaluation only.